Stage 4 Report: Diffusion and Project Progress

Image Super-Resolution with Deep Generative Models · Computer Vision 2026 · 27 Oct 2026 Team: Iuliia Kurnaeva, Alena Petrenko, Amaliya Kharisova, Dmitrii Kuznetsov

Repository: https://github.com/Ephy01/Image-Super-Resolution-with-Deep-Generative-Models

1. Completed experiments

All four methods were evaluated on 64 dev images (20 objects, 20 faces, 24 text), with JPEG none/50/30/10 and ×4 upscaling (128×128 to 512×512): **256 inputs per method**. No training or fine-tuning.

- **Diffusion.** Official **ResShift v3** and autoencoder; pinned implementation, four steps, batch 1, Tesla T4, float32 weights/float16 autocast. Fixed sampling seed per image, shared across JPEG levels; global/bootstrap seed 42. Mean **0.909 s/image** after three warm-ups; baseline timings from another GPU are not comparable. No Stable Diffusion fallback needed.
- **Reproducibility.** Only ResShift runs; archived inputs and baseline tables are reused. SHA-256 matches for all 336 archived files; all 320 images match local prepared inputs. Weights, versions, environment and timing are saved; outputs reproduce PSNR/SSIM. Cross-environment regeneration is unverified.

2. Preliminary comparison (dev split, 64 images)

PSNR/SSIM: Y channel, 4-pixel border crop. LPIPS: AlexNet. Values are image means; best mean per content/condition is bold.

Content	Method	PSNR ↑, no JPEG	SSIM ↑	LPIPS ↓	PSNR ↑, JPEG 10	SSIM ↑	LPIPS ↓
Objects	Bicubic	25.62	0.689	0.488	22.77	0.543	0.767
Objects	Real-ESRNet	**26.31**	**0.737**	0.376	**22.97**	**0.582**	0.613
Objects	Real-ESRGAN	24.44	0.682	0.255	21.80	0.547	**0.451**
Objects	ResShift	25.47	0.701	**0.212**	22.26	0.528	0.456
Faces	Bicubic	33.40	0.879	0.289	28.25	0.768	0.647
Faces	Real-ESRNet	**33.76**	**0.894**	0.238	**28.72**	**0.802**	0.456
Faces	Real-ESRGAN	31.05	0.859	0.159	27.65	0.776	0.342
Faces	ResShift	32.26	0.862	**0.108**	28.01	0.761	**0.322**
Text	Bicubic	16.46	0.707	0.436	15.12	0.581	0.639
Text	Real-ESRNet	19.31	0.838	0.186	**15.67**	**0.690**	0.408
Text	Real-ESRGAN	17.83	0.818	0.107	14.38	0.680	0.212
Text	ResShift	**19.76**	**0.858**	**0.085**	14.96	0.669	**0.195**

Paired PSNR differences without JPEG (dB). Brackets give 95% percentile bootstrap intervals: 10,000 resamples of within-image differences, seed 42; n = 20/20/24. Intervals are pointwise, without multiple-comparison correction.

Content	ResShift - Real-ESRNet	ResShift - Real-ESRGAN
Objects	-0.84 [-1.11, -0.57]	+1.03 [0.75, 1.34]
Faces	-1.50 [-1.85, -1.18]	+1.21 [0.85, 1.59]
Text	+0.45 [0.22, 0.67]	+1.92 [1.72, 2.14]

3. Observations and limitations

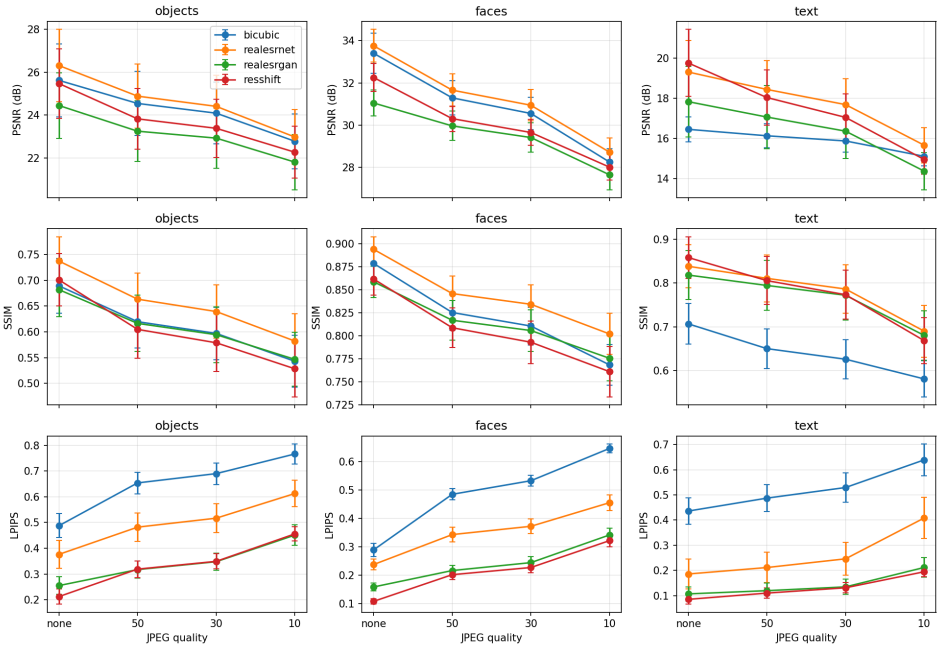
- **Text is an exception to the PSNR hypothesis.** Without JPEG, ResShift beats Real-ESRNet in PSNR and SSIM (SSIM difference +0.0199 [0.0157, 0.0243]). PSNR differences by HR font size: 16 px: -0.29 [-0.48, -0.13]; 24 px: +0.39 [0.20, 0.58]; 32 px: +1.00 [0.76, 1.33]; 48 px: +0.70 [0.42, 0.96] dB. These exploratory groups have six images each; the smallest font loses PSNR.
- **JPEG 10 repeats the GAN finding.** ResShift falls below bicubic in mean PSNR: objects -0.51 [-0.81, -0.25], faces -0.24 [-0.43, -0.06], text -0.16 [-0.44, 0.16] dB; the text interval includes zero. Against Real-ESRGAN, PSNR rises but SSIM falls in all categories. JPEG 10 is outside the configured [30, 95] training range, but this does not prove causation.
- **Simultaneous mean improvements.** Without JPEG, ResShift improves PSNR, SSIM and LPIPS over Real-ESRGAN in every category. This is not a universal rejection of the perception-distortion trade-off: the face SSIM difference is +0.0030 [-0.0021, 0.0082]. Lower LPIPS also does not establish correct letters or preserved identity.
- **Comparison limits.** Architecture, training, autoencoder, sampling and inference precision differ; gains cannot be attributed to diffusion alone. The journal paper, §IV-A names ImageNet for SR training, but the pinned v3 configuration also lists FFHQ. Overlap with our FFHQ images cannot be excluded; image-level membership is unverified.

4. Remaining work

- Add MUSIQ, OCR error rates by font size, and ArcFace similarity and match rate, with the threshold selected on dev.
- Analyse at least three failure cases with likely causes; produce the fidelity-versus-perceptual-quality plot and assess the JPEG-below-30 hypothesis.
- Repeat ResShift across seeds; benchmark all learned methods on the same GPU, including peak memory.
- Verify a complete four-method reproduction workflow, README and dependencies; report percentile interval endpoints consistently and complete the final evaluation with fixed settings.
- Prepare the two-page final summary, individual contributions, and a five-minute demo with at most three slides.

5. Figures

JPEG sensitivity. Existing bars show mean ± half the 95% bootstrap percentile span (10,000 resamples); they need not equal the percentile endpoints. Numerical intervals above report the endpoints directly.



ResShift at JPEG 10: low-resolution input, reconstruction and reference. Full images; sharper strokes do not establish correct characters, and facial details can change.

